# Social Context-awareness

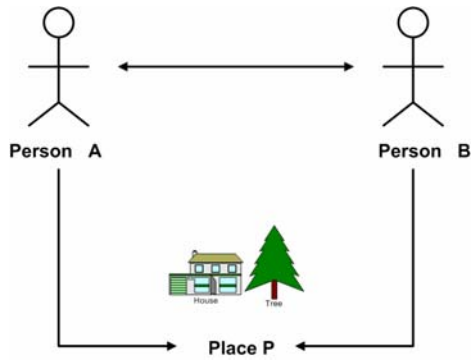
Ing. Gabriella Castelli
gabriella.castelli@unimore.it

# Outline

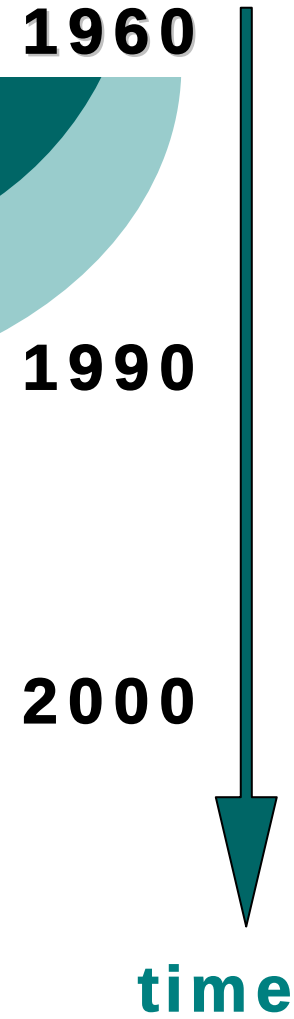
- P3 Systems
- Some examples
- Place Discovery, Labeling, and Representation
- Location based services:
  - Recommender Systems
  - Geotemporal social matching
- Privacy
- Browsing the World
- Google Earth

# Part 1

## P3 Systems



# The Interaction Society




**Calculation**: the computer of yesterday was occupied with crunching numbers

**Communication**: improvements in communication people-to-people to irrespective of local geography (Email, ICQ, Skype, etc.)

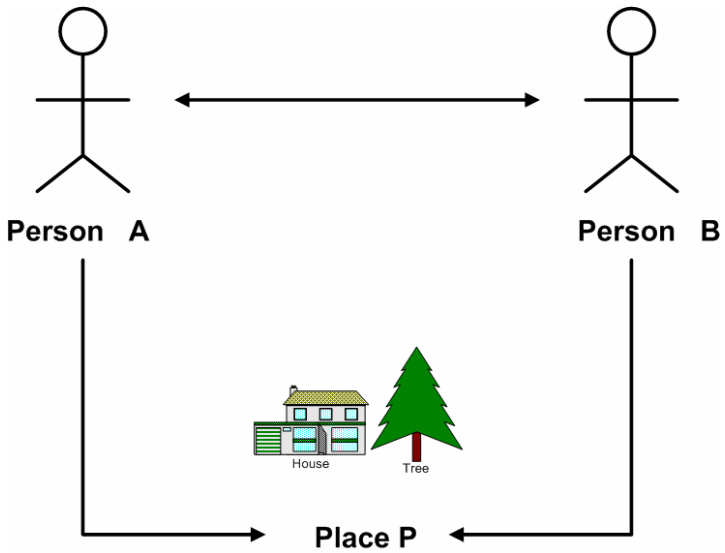
**Interaction**: location aware computing systems can locate individuals as they go about their daily activities and help their social life.

# Enabling Technologies

- **Portable devices**: mobile phones, palms, notebook
- **Communication Infrastructure**: wi-fi, umts, bluetooth, rfid, sensor nets
- **Localization technologies**: GPS, e-odt, a-gps

# People-to-people-to-geographical-places Systems

The availability of such technologies enables a new class of location-aware systems that link **people-to-people-to-geographical-places (P3 systems).** P3 systems combine computer-mediated communication and location data to provide appropriate geographic context to person-to-person interaction.




- P3 systems can strenghen the relationship between social networks and physical places.
- They can also help individuals leverage location information to make new social ties and coordinate interaction that reinforces existing ones.

# Contextual Factors of a P3 Systems

P3 Systems take into account:

- **People properties**: individual user's attributes and properties, including general attitudes as well as interests and current disposition
- **Place properties**: include place-type (such as restaurant or classroom), who frequents the location in question, social norm concerning people's expected behaviours in that place, etc.
- **People-place relationship**: include user's familiarity with a given place and details such as whether they have distinct roles in this location
- **People-people relationship**:include whether users already know each other, have mutual acquaitance, belong to the same organization, etc.

# P3 Systems design approaches

| **User interface** | | **Synchronous communication** | **Asynchronous communication** |
|---|---|---|---|
| People-centered | (1)Absolute user location | Uses remote awareness of current user location | Uses people location histories |
| | (2) Co-proximity | Uses real-time inter-users co-location for the exchage of social information | Uses co-location history to enable future interaction |
| Place-centerd | (3) Use of physical places by people | Uses online representation of user's current use of physical spaces | Uses history of people's use of a particular space |
| | (4) Interaction in Matching Virtual Places | Uses synchronous online interaction spaces related to physical location | Use asynchronous online places interaction related to physical location |

# People-centered Vs Place-centered

- **People-centered** tecniques use location information to support interpersonal awareness, enable communication, etc.
  - Eg., The Weasley clock in Harry Potter




- **Place-centered** tecniques link virtual places to physical locations, using social information to aid place-based navigation and decision making
  - E.g., The Marauder's Map in Harry Potter



# Absolute location Vs Co-proximity

- **Absolute user locations** techniques tell you where your buddies are (latitude, longitude, altitude).
  E.g., (44.00, 10.33, 100), at home, room 35

- **Proximity** between users –or relative locations- techniques only tell which buddies are close to you or to a friend.

# Synchronous Vs Asynchronous Communication

- **Synchronous location awareness** refers to the provision of current information about user location. Note: this location awareness need not necessarily be reciprocal, in the sense that the system may provide a user with a buddy's location without necessarily providing the buddy with the user's location.

- **Asynchronous location awareness** refers to the provision of historical information about user location.

Part 2

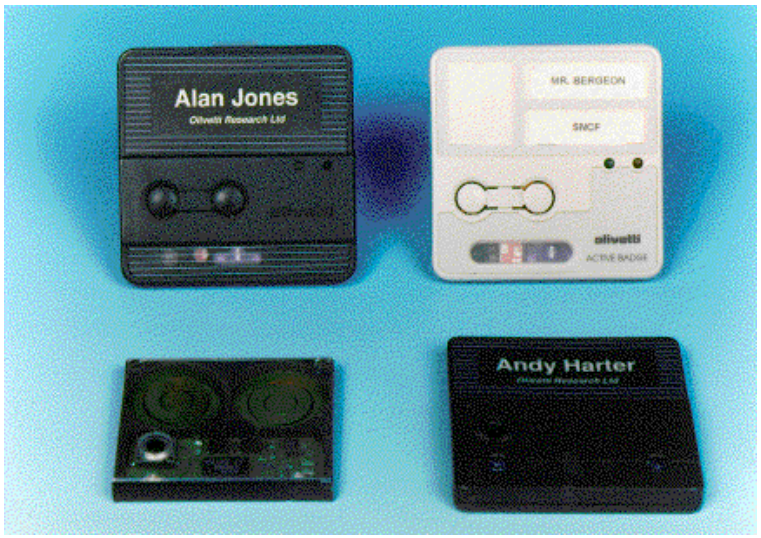
# P3 Systems Examples

# Representative P3 Systems

| **User interface** | | **Synchronous communication** | **Asynchronous communication** |
|---|---|---|---|
| People-centered | (1)Absolute user location | ActiveBadge, Ulocate | Ulocate's history function |
| | (2) Co-proximity | LoveGety | SocialNet |
| Place-centerd | (3) Use of physical places by people | ActiveCampus | Folkmusic |
| | (4) Interaction in Matching Virtual Places | Geonotes | ActiveCampus |

# Active Badge

The Active Badge system provides a means of locating individuals within a building by determining the location of their Active Badge. This small device worn by personnel transmits a unique **infra-red signal** every 10 seconds. Each office within a building is equipped with one or more **networked sensors** which detect these transmissions.




The location of the badge (and hence its wearer) can thus be determined on the basis of information provided by these sensors.

# Ulocate

**uLocate** is North America's leading publisher of mobile location services. As a pioneer in the mobile LBS market, uLocate have launched applications including friend finders, point of interest locators, and photo tagging applications.



# LoveGety

The Japanese LoveGety supports human interaction across really short distances. The LoveGety is a **matchmaking device** that communicates with **co-located devices** in the near proximity of its user, i.e., around 10 meters.

- There's a LoveGety for man (blue underside), and a LoveGety for woman (pink underside)
  - Owners can set the device to show display lights according to whatever mood they are in (there are only three): "let's just chat," "let's go sing some karaoke," or the "Get together" mode.
  - When one LoveGety detects another of the opposite sex within range, it beeps and flashes green if both are in the same harmonious mode, and red if the opposite user is sending out a different mode.



# ActiveCampus

The ActiveCampus project aims to provide location-based services for educational networks.

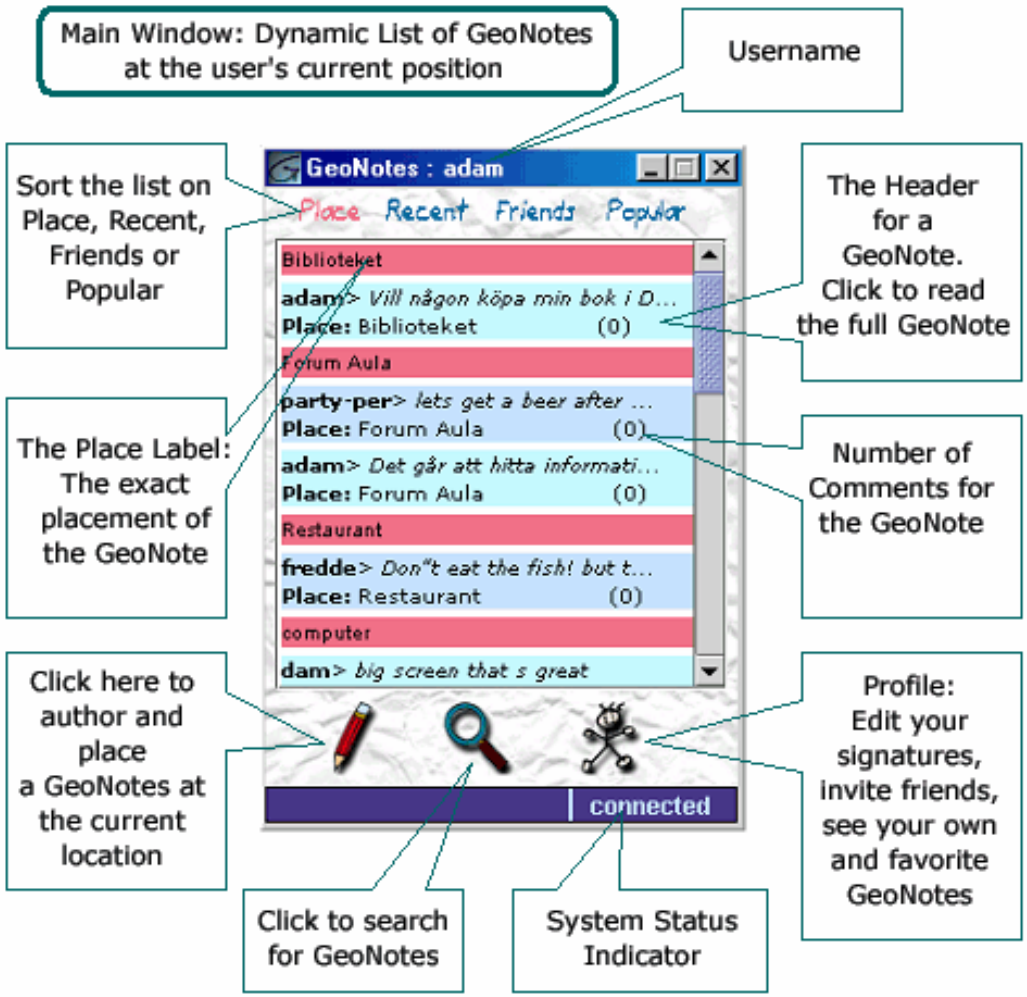
Two services:

- **ActiveClass** enables collaboration between students and professors by serving as a visual moderator for classroom interaction. E.g., Ask anonymous questions. (Class as Virtual Place)



- **ActiveCampus Explorer** uses a person's context, like location, to help engage them in campus life. It consists in a suite of complementary services:
  - What's around
  - Who's around
  - Gaffiti (Msn, ICQ, etc…)

# Geonotes




When you want to create a **GeoNote** at your present location, you do not have to specify your position to the system. Via your connection to the Wireless LAN base-station (see Before you start the application) , GeoNotes automatically detects where you are. You are always 'here'.

However, W-LAN positioning can be rather rough and crude. In order to give GeoNotes authors and readers guidance to its exact position, each GeoNote is filed under a place label. **Place labels** are created by authors each time they create a GeoNote.

# Part 3

## Place Discovery, Labeling, and Representation

# Place Discovery, Labeling, and Representation

P3 Systems need to represent places, obtain description for them and identify when a user is ‘‘at’’ a place. Algorithms for defining places and obtain labels are not stated.

Some guide lines:

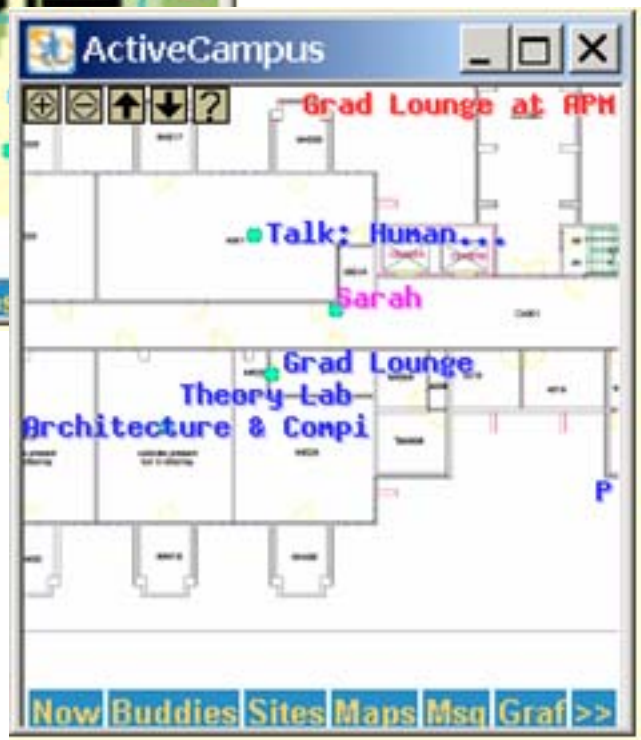
- **Expressive representations of places**, combine geometric representation (e.g., latitude,longitude) and logical representations based on places, their interrelationships, and perhaps even their temporal properties (e.g., a room might be used as a class room during the day and dance floor in the evening.)
- Algorithms for aiding users in social navigation tasks must also be able to utilize **relational and dynamic properties of places**.

  For example: What does it mean for a place to be crowded?
  When is discourse typically active?

- Scalability requires algorithms to be interactive and place **descriptions to be shared among users**

# Different Places Representation






# Part 4

## Location Based Services

- P3 Recommender Systems
- Geotemporal Social Matching

# Location Based Services

- LBS provide mobile-device with personalized services tailored to their locations.
  - Reactive LBS
  - Proactive LBS

- Classification of time and space resolution for information:

| | **Geographically located** | **Geographically indipendent** |
|---|---|---|
| **Localized in time** | Traffic reports | Share price |
| **Time indipendent** | Restaurant location | Music recordings |

# Location Based Services (2)

Desire for geographically located information:

| | **Stable information** | **Dynamic Information** |
|---|---|---|
| **Activity done in a place frequently** | **Need: Low**<br>Eg., What is the train schedule? | **Need:Moderate/High**<br>Is the 10:17 a.m.train on time today? |
| **Activity done in a place infrequently** | **Need: Moderate/High**<br>How do I get to a restaurant I rarely visit? | **Need: High**<br>What movies are playing this afternoon at a cinema I don't go often? |

## P3 LBS:

- P3 Recommender Systems
- Geotemporal Social Matching

# P3 Recommender Systems

- **Recommender systems** use knowledge of user preferences and item properties to identify what users are likely to enjoy.

  E.g. news services on cell phones

- **A P3 recommender system** represents and resons with user preferences – either inferred from use or stated explicitly – about both places and people activities.

  E.g. A system might recommend natural foods restaurant to users that regularly visit organic markets.

# P3 Recommender Systems (2)

- We can use **collaborative techniques** to derive more effective reccomendation.

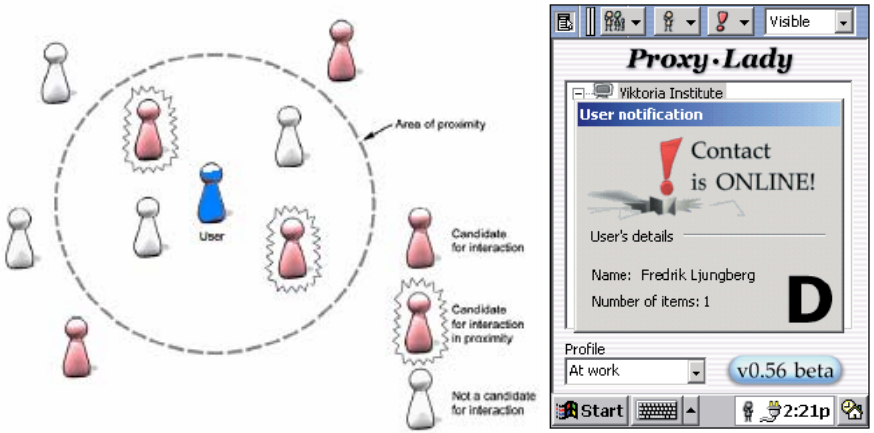
  E.g., A system could infer users' preferences for particular types of cousine from the restaurants they frequent.

  The system might base restaurant recommendation combining users' preferences and restaurant's current status (wait time for a table, ambient noise levele, etc.)

- When virtual spaces are associated with physical places, a common use is to let users enter **ratings and** express **opinions**, which place-based recommender systems could use in turn.

# Geotemporal Social Matching

**Social matching systems** are recommender systems that bring people together in both physical and online spaces.





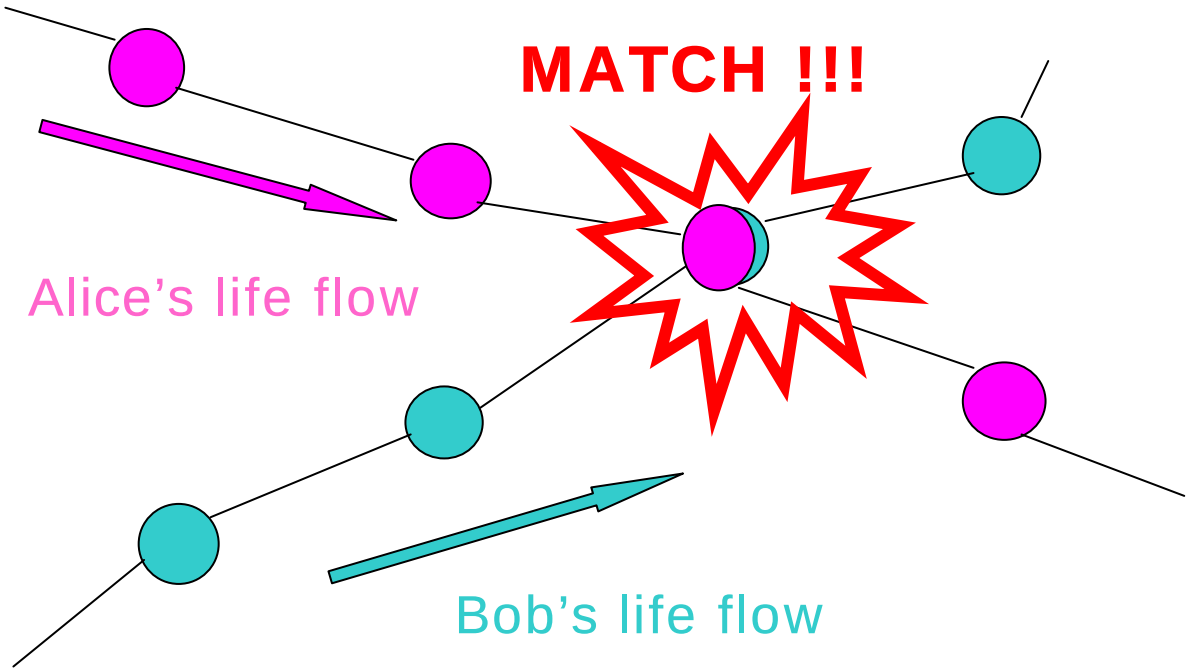


- Social matching systems typically include **match-alert mechanisms** and introduction-management tools.
- They often do so by providing users with access to various aspects of other **users' profiles** through listings or social network visualizations. Profiles can change over time and be linked to **reputation measures**.

# Geotemporal Social Matching (2)

**Geotemporal social matching** systems are a class of P3 recommender systems that leverage users' geotemporal histories to match individuals.

In everyday life, we often use location to find social matches without the aid of computers — for example, we find our colleagues at work, and we often meet friends at pubs…




**Same time**

**&**

**Same place**

# Geotemporal Social Matching (3)

**Social-matching algorithms** exploit, and combine, each of these P3 system techniques:

- We can use stored **absolute-user-location** data to derive affinities based on similarities among users' geotemporal routines. Algorithms for matching location histories search for common or nearby locations and similar paths. Such algorithms would be even more effective if they learned to use information about place-types;
- **Collocation data** is the most common trigger for geotemporal match alerts: when individuals with matching profiles come into proximity, they receive alerts suggesting that they meet. We need to combine such introduction alerts with identity-revelation tools that let users meet while maintaining control of their personal data.
- Use of **physical places data** can be used to determine the appropriateness in terms of timing and content of individual match alerts.
- Users' use of **matching virtual places** provide a safe place for virtual, and perhaps asynchronous, introductions.

# Part 5

## Privacy Concerns

# Privacy Concerns

Many researchers have explored **privacy and security issues** in collaborative and ubiquitous computing systems.

The P3 systems framework can help us identify the privacy concerns associated with the various techniques:

- For **absolute user location techniques**, the key issue of concern is the possibility of "stalking" or simple violations of users' desire for privacy. Using this technique makes sense only in the context of strong social ties between users e.g., family members.
- For **collocation proximity** techniques, the key concerns are associated with geotemporal social matching and identity management. E.g., an individual waiting @ a train platform
- Uses of **physical places** techniques don't necessarily raise as many privacy concerns as people-centered techniques because they often require only anonymous data about physical activities in a given place.
- The interactions in **matching virtual places** techniques raise privacy issues similar to more traditional forms of computer mediated communication systems.

# Part 6

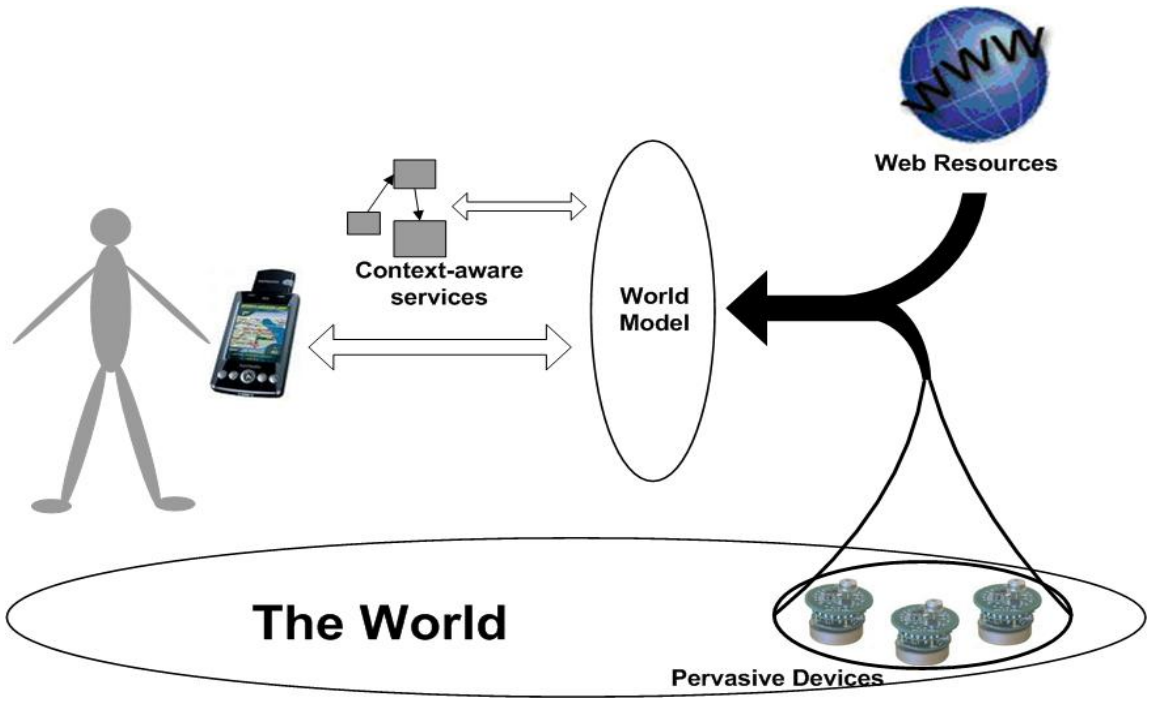
## Browsing the world

by Agent and Pervasive Computing Group

# Browsing the world
## Agent and Pervasive Computing Group

It's a **general model** to represent context information and to build a world model.

- We provide a **general infrastructure** supporting the model



# The W4 Context Model

- A simple model in which context data is expressed by a four field structure: Who, What, Where and When.
- Someone or something (***Who***) does some activity (***What***) in a certain place (***Where***) at a specific time (***When***)
- Who is acting? What is he/she/it doing? Where and when the action takes place?

# W4 examples

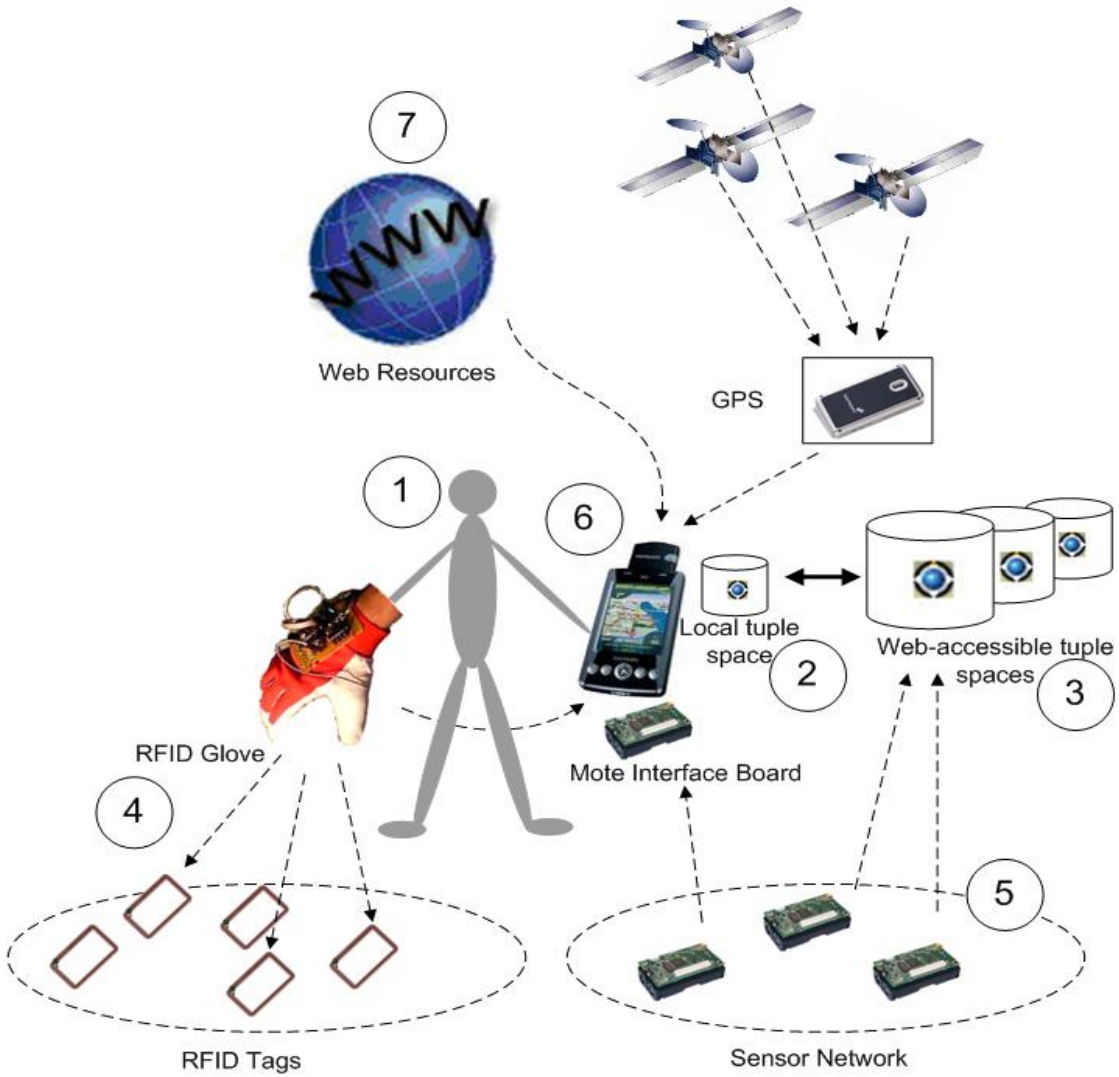
Gabriella is walking in the campus' park. An agent running on her PDA can periodically create a **knowledge atom** describing her situation.

**Who**: user:Gabriella
**What**: works:pervasive computing group
**Where**: lonY, latX
**When**: now

- Gabriella is walking in the campus, and wants to know if some colleague is near. She will ask:

**Who:** user:*
**What:** works:pervasive computing group
**Where**: circle,center(lonY,latX),radius:500m
**When:** now

# Browsing the World Architecture



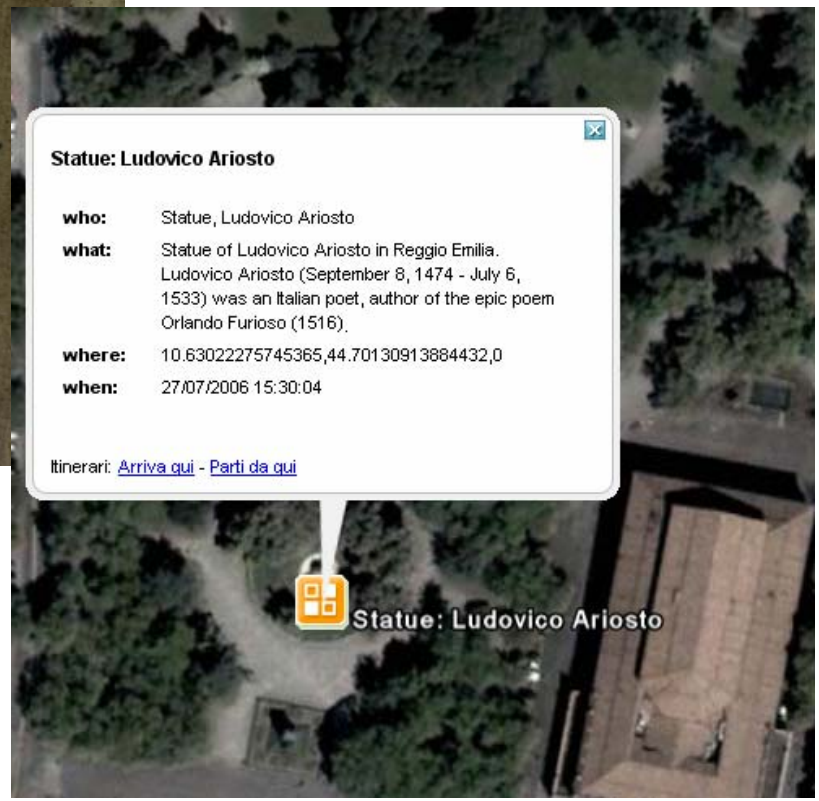
# Journey Map

A tourist wants to automatically build and maintain a diary of his journey.



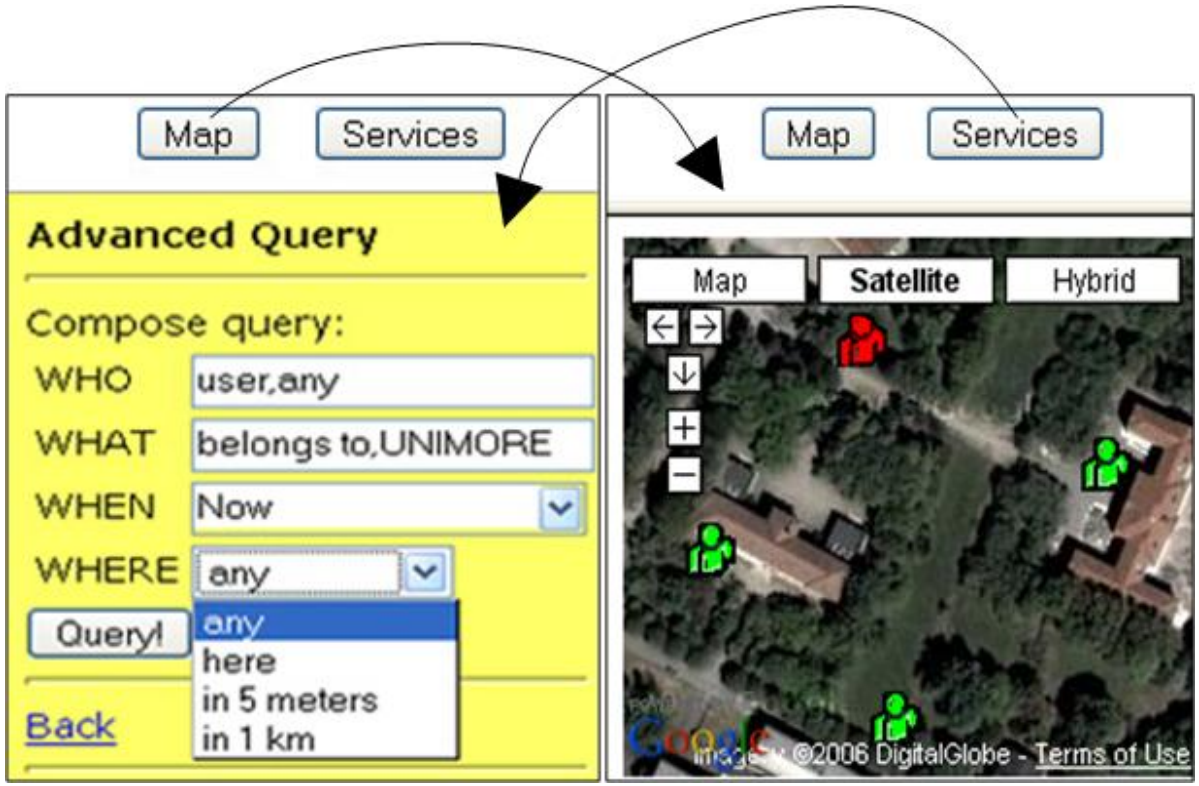
# Journey Map (2)



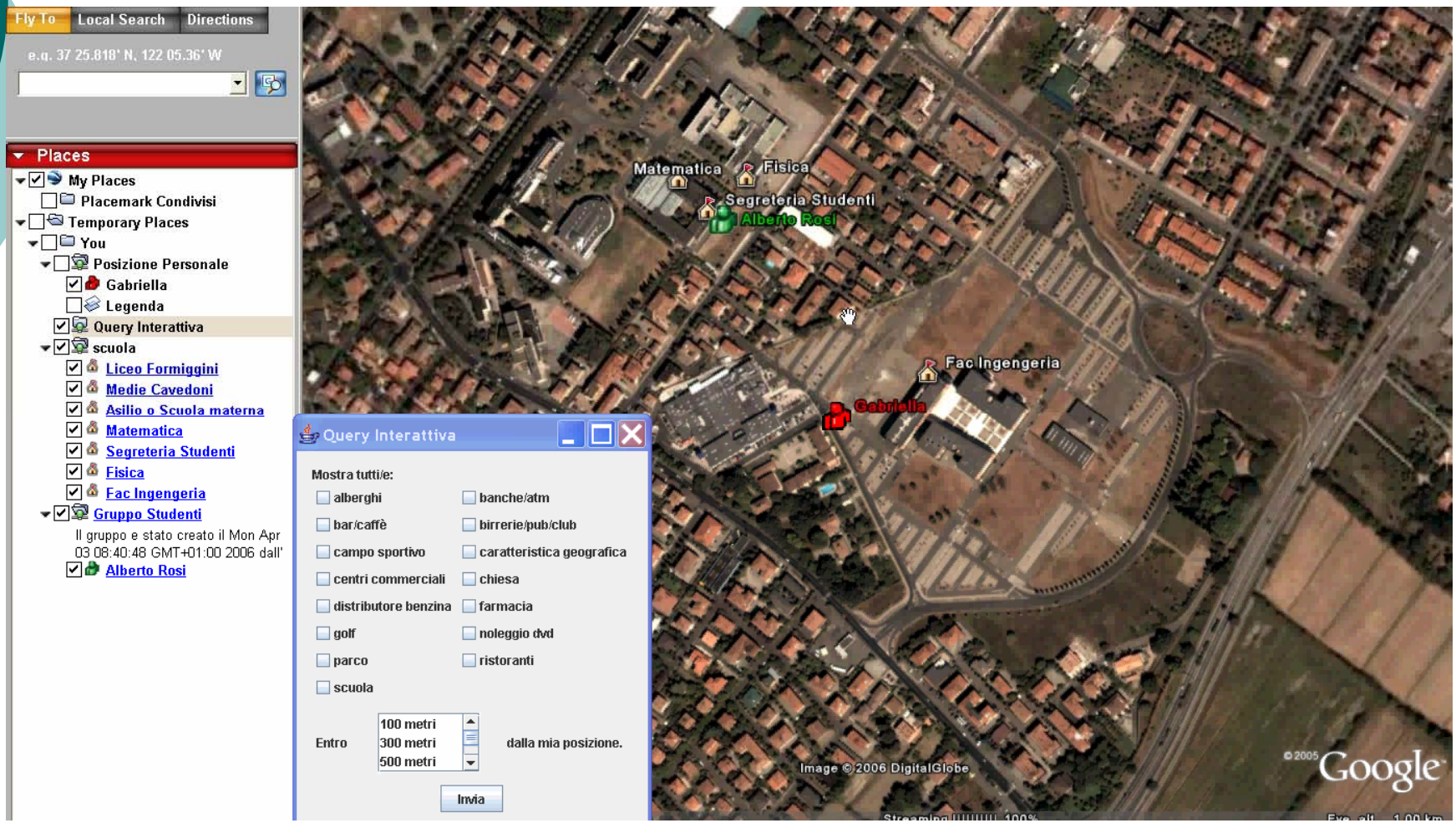


# The People Map

Group of friends can share their actual GPS locations (represented as knowledge atoms) with each other.



# Real Time Map



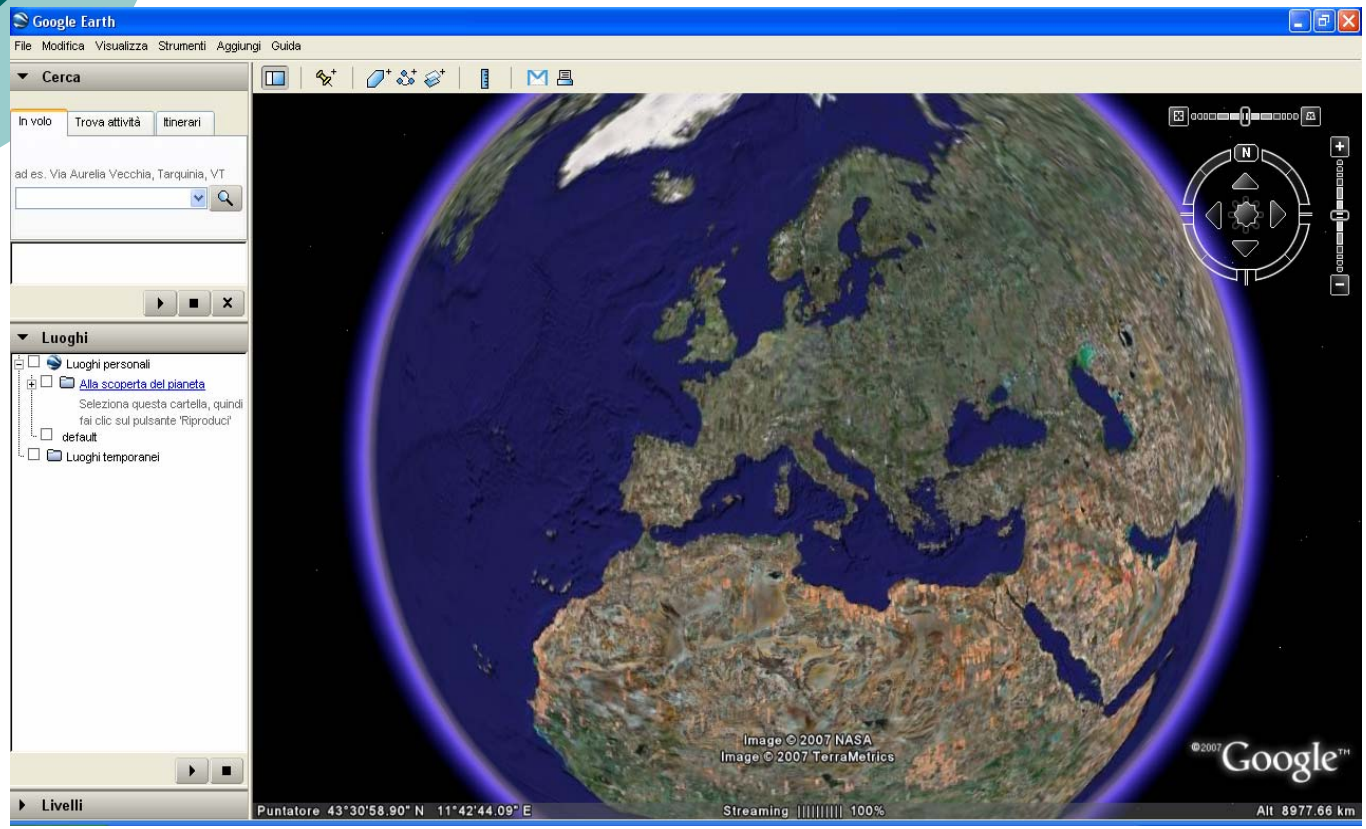
# Part 7

## Google Earth

making P3 possible…

# Google Earth

Google Earth combines the power of Google Search with satellite imagery, maps, terrain and 3D buildings to put the world's geographic information at your fingertips.




- Fly to your house.
- Search for schools, parks, restaurants, and hotels.
- Tilt and rotate the view to see 3D terrain and buildings.

# Motivation

**Google Earth** is an open software (i.e. it has public and free API). Users can continuously enrich GE data by geo-located information coming from facts and events situated in the world: point of interest, images, 3D objects etc.

GE allows also to specify the user viewpoint on the map. This naturally supports **context awareness**, in that you can decide to center the map where relevant information are located.

In particular, by centering the map on the user, it provides a **user-centric** representation of the world.

# Google Earth API: KML

**KML** (Keyhole Markup Language) is a file format used to display geographic data in an earth browser, such as Google Earth, Google Maps, and Google Maps for mobile.

You can use KML to:

- Specify icons and labels to identify locations on the planet surface
- Create different camera positions
- Use image overlays attached to the ground or screen
- Define styles to specify feature appearance
- Write HTML descriptions of features
- Use folders for hierarchical grouping of features
- Dynamically fetch and update KML files from remote or local network locations
- Fetch KML data based on changes in the 3D viewer
- Display COLLADA textured 3D objects

# KML rules

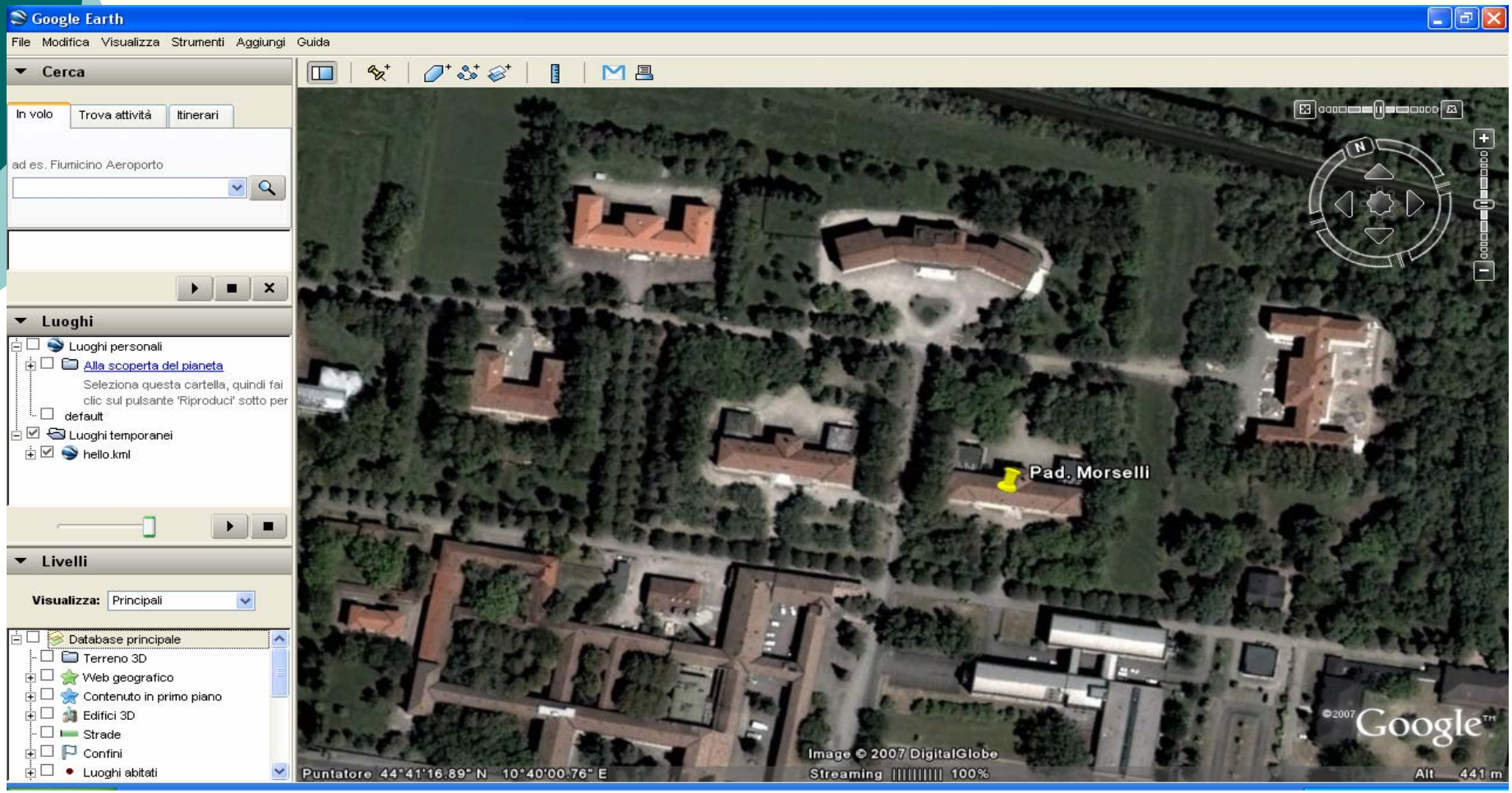
The structure of a KML document obeys all of the syntax **rules of XML**:

- XML tags must always have an opening and closing tag
- XML tags are case sensitive
- XML elements must be properly nested and well formed
- An XML document must always have a single root element
  For KML, this means you can use the <**kml**></**kml**> tag
- Attribute values must always be quoted
- CR/LF is converted to a new line

**KML tag types**:

- Complex tags are easily identified by an initial upper case letter and can function as parent tags to both complex and simple tags.
- simple tags use all lower case. They're are children only and can contain no other tags .

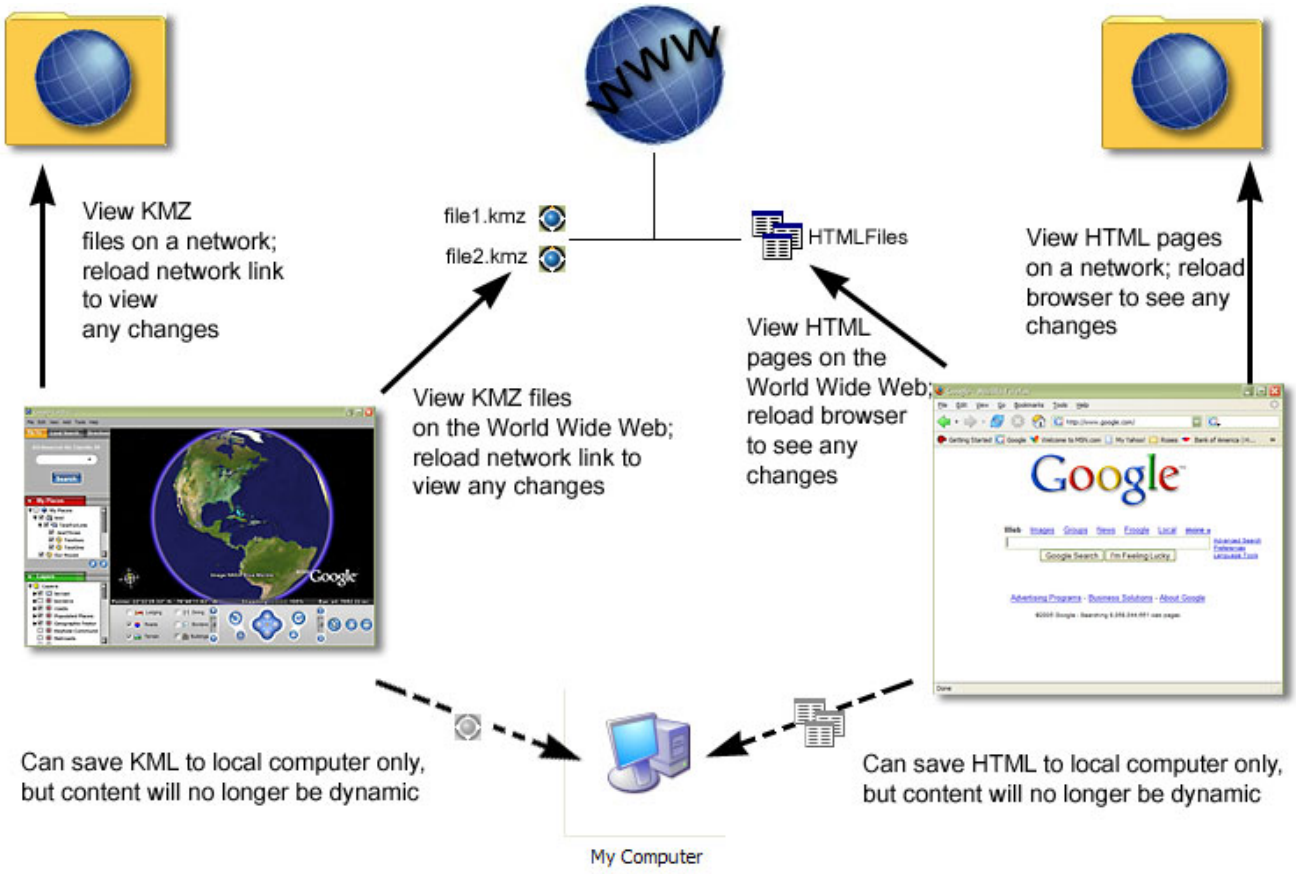
# Placemark


Google Earth
File Modifica Visualizza Strumenti Aggiungi Guida
Cerca
In volo
Trova attività
Itinerari
ad es. Fiumicino Aeroporto
Luoghi
Luoghi personali
Alla scoperta del pianeta
Seleziona questa cartella, quindi fai clic sul pulsante 'Riproduci' sotto per
default
Luoghi temporanei
hello.kml
Livelli
Visualizza: Principali
Database principale
Terreno 3D
Web geografico
Contenuto in primo piano
Edifici 3D
Strade
Confini
Luoghi abitati
Pad. Morselli
Image © 2007 DigitalGlobe
©2007 Google
Puntatore 44°41'16.89" N 10°40'00.76" E
Streaming 100%
Alt 441 m

# Placemark

```
<?xml version="1.0" encoding="UTF-8"?>
<kml xmlns="http://earth.google.com/kml/2.1">
    <Placemark>
        <name>Pad. Morselli</name>
        <description>Dipartimento di Scienze e Metodi
                dell&amp;apos;Ingegnria</description>
        <LookAt>
                <longitude>10.66687692615733</longitude>
                <latitude>44.68802532597002</latitude>
                <altitude>0</altitude>
                <range>440.9748132299762</range>
                <tilt>-1.218615474991631e-011</tilt>
                <heading>17.95420797374584</heading>
        </LookAt>
        <Point>
                <coordinates>10.66732473965654,
                        44.6877040302364,0</coordinates>
        </Point>
 </Placemark>
</kml>
```

# Network Link




**Network Links with Google Earth**

**Web Page Connection with Web Browser**

# NetworkLink.kml

```
<?xml version="1.0" encoding="UTF-8"?>
<kml xmlns="http://earth.google.com/kml/2.1">
<NetworkLink>
    <name>Morselli</name>
    <visibility>1</visibility>
    <open>1</open>
    <flyToView>1</flyToView>
    <Url>
        <href>http://localhost/unimore/esempio.jsp</href>
        <viewRefreshMode>onStop</viewRefreshMode>
        <viewRefreshTime>2</viewRefreshTime>
    </Url>
</NetworkLink>
</kml>
```

# Example.jsp

```
<?xml version="1.0" encoding="UTF-8"?>
<kml xmlns="http://earth.google.com/kml/2.1">
<%@ page language="java"%>
    <Placemark>
        <name>Pad. Morselli</name>
        <description>Date & time: <%= new
        java.util.Date()%> </description>
        <Point>
                <coordinates>10.66732473965654,
        44.6877040302364,0</coordinates>
        </Point>
   </Placemark>
</kml>
```

# Campus San Lazzaro

